Adhithya Laxman R G

SWE Intern · ETH Zurich MSc · Autonomous Vehicles · Computer Vision · 3D Reconstruction · Perception · DL
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Education

ETH Zurich (Swiss Federal Institute of Technology Zurich) — Zurich, Switzerland
MSc in Data Science *(CGPA : 5.1/6)* — Sep 2025 - Sep 2027
Relevant Coursework: Computer Vision, 3D Vision, Probabilistic AI, NLP, Deep Learning, Computational Statistics

Coimbatore Institute of Technology (Affiliated to Anna University) — Coimbatore, India
B.E. Computer Science and Engineering *CGPA: 9.57 (Top 2% of the class)* — Nov 2021 - May 2025

IIT Madras — Chennai, India
Diploma in Data Science & Applications (Online) *CGPA: 8.49* — Jan 2022 - Aug 2025

Experience

ETH Zurich - Computer Vision and Geometry Group | Research Student — Zurich | Feb 2026 - Jul 2026

- Improving robustness of **3D reconstruction pipeline** for in-the-wild indoor videos (HouseTour dataset, ICCV 2025) by integrating **feed-forward reconstruction models** to address **40% failure rate** in traditional SfM methods.
- Benchmarking and adapting **vision transformer models** (DUSt3R, MASt3R,DAV3) to challenging indoor video data; developed **hybrid SfM pipeline** with COLMAP backend, implementing **systematic evaluation pipelines** and validation workflows; leveraging multi-GPU training for ViT model benchmarking.

Caterpillar Inc. | Software Engineer Intern (Computer Vision / ML Systems) — Chennai | Jan 2025 - Jul 2025

- Developed and deployed **monocular depth estimation pipeline** for **autonomous truck perception** — training on 50k+ images in PyTorch, optimizing for **edge AI hardware** with **camera-only sensor constraints**.
- Optimized model for edge hardware, reducing depth perception error to 4% and inference time by 10 ms for resource-constrained, enabling real-time inference in camera-only systems.
- Applied **MLOps practices** (Git, Docker, Azure DevOps) to automate training, deployment, and ensure reusability.

Akamai Technologies | Software Engineer Summer Intern (DevOps) — Bangalore | Jun 2024 - Aug 2024

- Built automated **ETL Data pipelines** using **Jenkins** to process large-scale data from **Hive clusters** with over **1 billion records daily**. Integrated **ClickHouse** with **Grafana dashboards** for real-time KPI visualization.
- Migrated **6 legacy pipeline components** (Python, ClickHouse, Jenkins jobs) to current versions, reducing **pipeline failures** and improving **deployment reliability** across the Release DevOps team.
- Collaborated with DevOps and data engineering teams in **agile environment**, delivered features in tight deadlines.

Samsung R&D Institute | Research Intern (Samsung PRISM) — Aug 2023 - Mar 2024

- Led a team of 5 in the development of **Conditional GAN models** in TensorFlow to generate MNIST-based video sequences, improving accuracy by 15% with **custom pix2pix loss functions** achieving **0.97 SSIM** accuracy.

Skills

Languages:	Python, C++, Java, C, SQL, Shell Script
ML & Frameworks:	PyTorch, TensorFlow, ONNX, Scikit-learn, NumPy, Pandas, Flask, Django
Focus Areas:	Deep Learning, Computer Vision, Edge AI, NLP, RAG, Multimodal (Vision-Language), Transformers, Distributed Systems
Data & Tools:	AWS S3, GCP, Kubernetes, Hive, Spark, ClickHouse, MongoDB, PostgreSQL, Docker, Git, Jenkins, Azure DevOps, Linux, Grafana

Machine Learning Projects / Open-Source

Investigating Relational Reasoning in Vision Language Models | Link

- **PyTorch, Qwen 3-4B, Mechanistic Interpretability, CLEVR, ChartQA**
- Analyzing attention mechanisms in **Qwen 3-4B VLM** across **4 synthetic datasets** using mechanistic interpretability **(ablation, activation patching, circuit analysis)** to identify relational reasoning limitations and knowledge shortcuts in LLM behavior.
- Evaluating **cross-modal attention patterns** using **IoU** and mutual information metrics to quantify visual-text alignment; applying **Linear Probing** across layers and heads to identify shortcut learnings in spatial reasoning.

Federated IDS with Deep Belief Networks | Link

- Led team of 5 developing federated learning system for time-series sensor data classification (WiFi network signals), implementing privacy-preserving distributed training on 300k+ data using PyTorch and custom gRPC architecture.
- Designed robust data pipelines for heterogeneous sensor networks, achieving 98% accuracy through systematic benchmarking of Deep Belief Network architectures and automated validation workflows.

Honors & Awards

- Accepted paper at **Springer 2025: FEDDBN-IDS** Federated Deep Belief Network for Intrusion Detection on Heterogeneous WiFi Networks for previously unknown attacks. Link
- Winner, **Internal Smart India Hackathon 2023**(1/16) ;2nd place **Innovision24 Hackathon** (2/48 teams)
- **"A. John Wilson Endowment Prize"** for the highest mark in **Mathematics** across departments
- **Leetcode Knight** Top 2.5% Globally (Rating 2020+)
- Awarded "Excellence Certificate" for performance and complex implementation in Samsung PRISM internship
- **Made open-source contributions** to TheAlgorithms/Python (180k+ stars) adding neural network algorithms and optimization techniques